

EC-201: Engineering Project Management

Broad Contents

- Management
- Key management concepts
- Functions of management
- Comparison of 20th and 21st century organizations

1.1 What is Management?

- **Managing** is an art of getting things done through and with people in formally organized groups.
- **Management** is the process of designing and maintaining an environment in which individuals, working together in groups, efficiently accomplish selected aims towards any project. It is the art of creating an environment in which people can perform as individuals and yet cooperate towards the attainment of group goals.
 - i. **Management as a Process:** According to this, management is the process of using organizational resources to achieve the organization's goals through planning, organizing, leading, and controlling. It is thus, a set of activities directed at an organization's resources with the aim of achieving organizational goals in an efficient and effective manner.
 - ii. **Management as People:** This refers to a group of people who engage in "Process of Management".

1.2 Key Management Concepts

- **Project Organization:** It comprises of people working together and coordinating their actions to achieve specific goals.
- **Goal:** A desired future condition that the organization seeks to achieve.
- **Resource:** An asset, competency, process, skill, or knowledge controlled by organization. Various types of resources that an organization possesses are as follows:
 - People
 - Information
 - Machinery
 - Financial capital
 - Raw Materials
- A resource is strength, if it provides an organization with a competitive advantage. On the contrary, a resource is a weakness; if it is something the organization does poorly or does not have capacity to do. Organizational resources include: Human, Physical, Financial, Technological, and Information.

1.3 Evolution of Management Concept in Modern Era (1/2)

a) *Frederick Taylor – Father of Scientific Management* was a Mechanical Engineer. He invented high speed steel cutting tools. He got the opportunity to know first hand problems and attitudes of the workers. Based on these he identified that in order to improve the quality of management, the major concern was to *increase efficiency in production, lower cost, raise profits through higher productivity, and also increase the pays/salaries of the workforce*. His message of management was to give people their best opportunities to be productive, and in turn reward workers for their individual productivity. This increase in labour productivity is not possible without the following:

- Providing ample rewards
- Adequate trainings
- Continuous managerial support

Thus, Fredrick Taylor concluded that *“low productivity in any project is matter of ignorance on part of labour and management”*.

1.3 Evolution of Management Concept in Modern Era (2/2)

b) **Henry L. Gantt** stressed the importance of *“developing understanding of systems both for labour as well as management.”* He emphasized that in all problems of management, human element is the most important one. Gantt gave graphic methods of describing project plans in order to have better managerial control. He highlighted the importance of time and cost in planning and controlling projects. He made the famous **Gantt chart**.

1.4 Key Aspects of the Management Process

- The key aspects of the Management Process can be explained with the help of the following diagram:

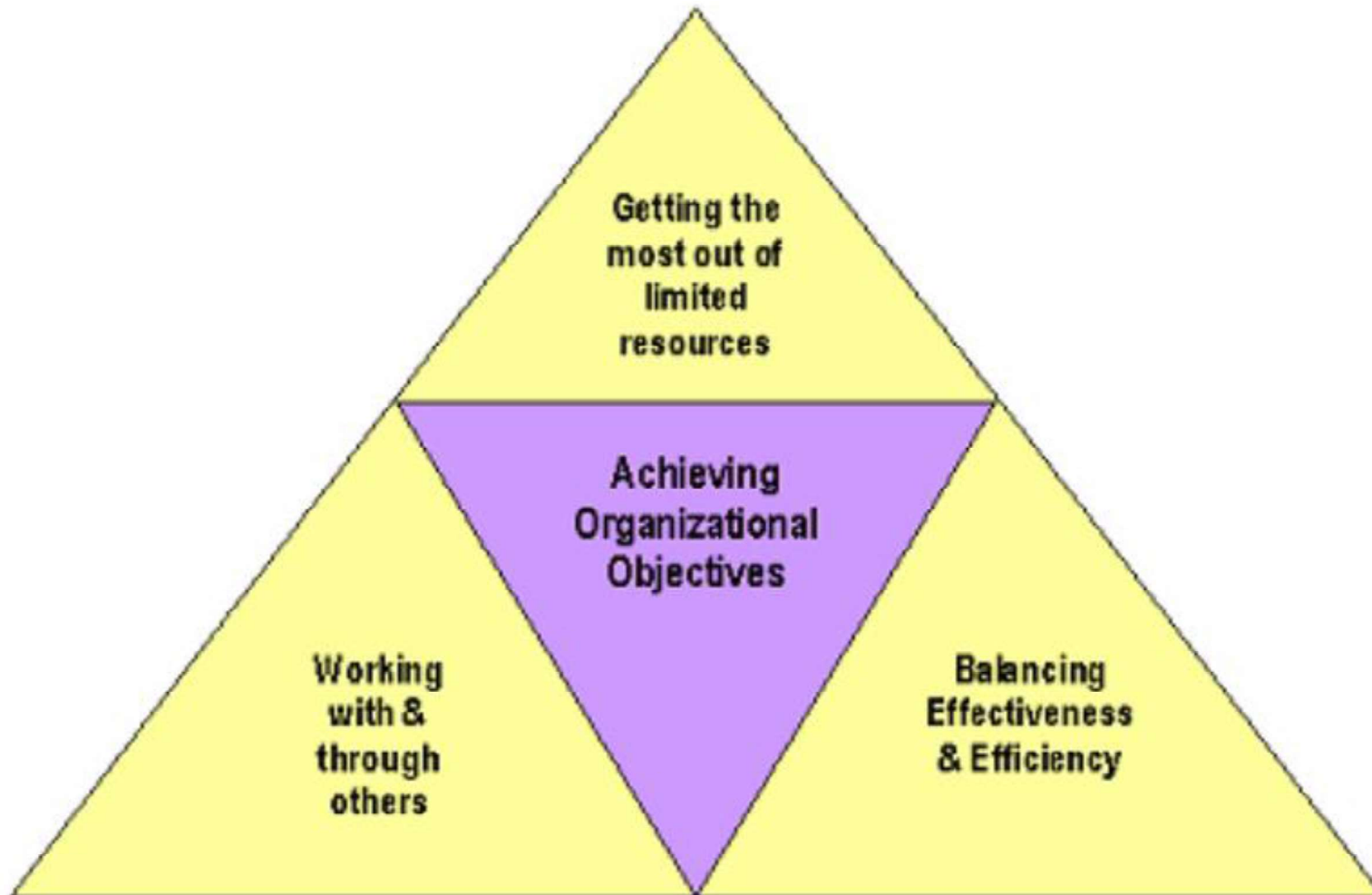


Figure 1.1: Management Process Aspects

1.5 Functions of Management (1 / 2)

The process of management consists of four basic managerial functions. These are:

1. **Planning** is the process of setting objectives in any project and then determining what should be done to accomplish them. It is a capstone activity of management. Managers at every level do planning. Planning activities determine an organization's objective and based on these helps it in establishing appropriate strategies for achieving them. The strategies are intended to achieve a sustained competitive advantage over the competitors.
2. **Organizing** is the process of assigning tasks, allocating resources, and arranging coordinated activities to implement plans. It involves establishing intentional structure of roles for people to fill in organizations.

1.5 Functions of Management (2/2)

3. **Leading:** is the process of bringing out enthusiasm and directing human resource efforts toward project and organizational goals. It involves influencing people so that they contribute towards organizational and group goals. Leadership predominantly is concerned with the interpersonal aspect of managing.

In projects most important problems arise from people in terms of their desires, attitudes, behaviour (as individuals as well as in groups). Thus, effective project managers also need to be effective leaders.

Leadership implies follower-ship and people tend to follow those who offer means of satisfying their own needs, wishes, and desires.

4. **Controlling:** Controlling is the process of measuring performance and taking actions to ensure desired results in any project. It involves measuring and correcting individual as well as organizational performance to ensure that events conform to plans.

1.6 Management Functions: Planning, Organizing, Leading & Controlling

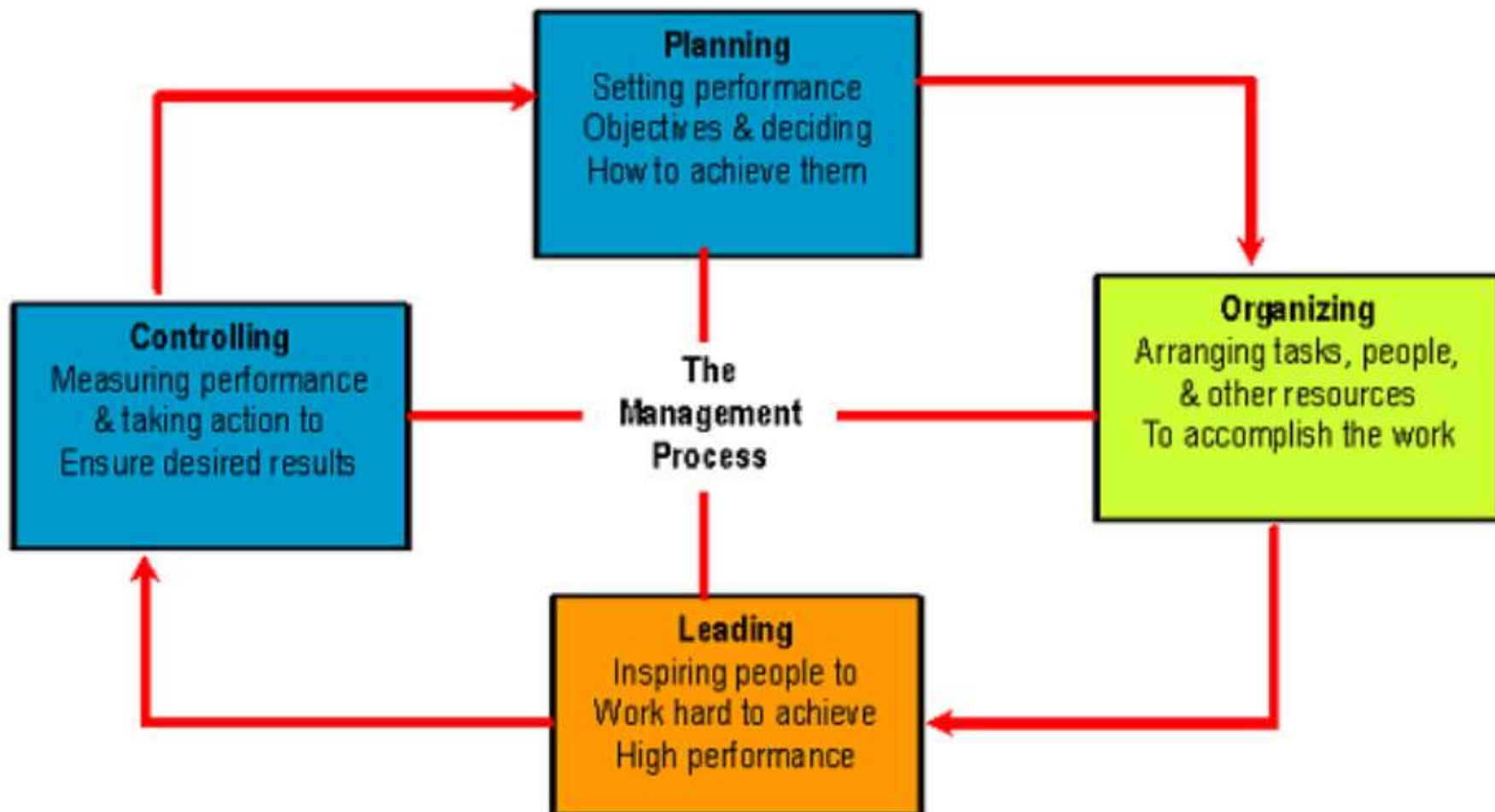


Figure 1.2: Management Functions

1.7 Managerial Functions in Organizations Undertaking Projects (1 / 2)

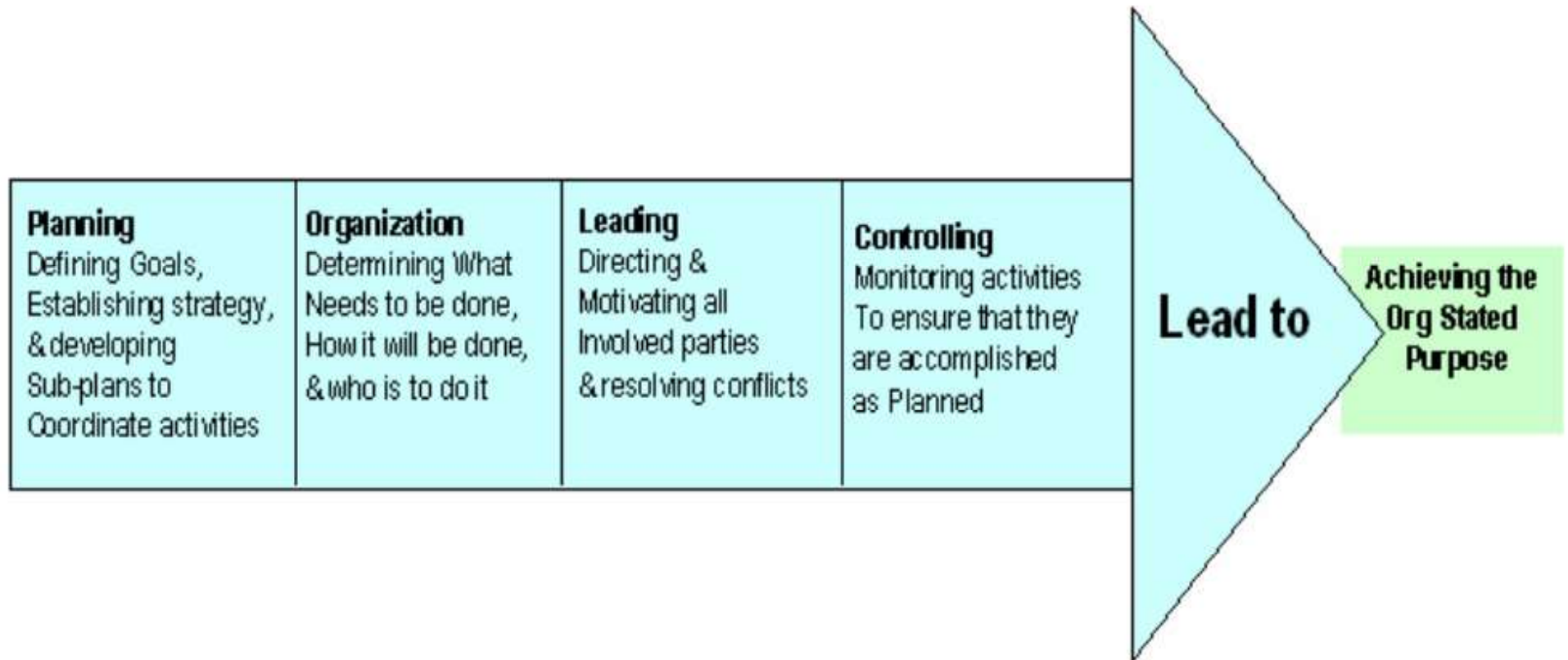


Figure 1.3: Managerial Functions

1.7 Managerial Functions in Organizations Undertaking Projects (2/2)

- Organizations are arranged in ways that try to maximize *synergy*, i.e. *the ability of the whole to equal more than the sum of its parts*. This means that an organization ought to be able to achieve its goals more effectively and efficiently than would be possible if the parts operated separately. Organizations comprise of various levels. These are depicted in the following figure:



Figure 1.4: Various organizational levels

1.8 Comparison of 20th and 21st Century Organizations

20 th Century Organizations	21 st Century Organizations
<u>Structure</u> • Bureaucratic • Multi-leveled • Organized with the expectation that senior management will manage • Characterized by policies and procedures that create many complicated internal interdependencies	<u>Structure</u> • Not bureaucratic, with fewer rules and employees • Limited to fewer levels • Organized with the expectation that management will lead, and lower-level employees will manage. • Characterized by policies and procedures that produce the minimal internal interdependence needed to serve customers.
<u>Systems</u> • Depend on few performance information systems. • Distribute performance data to executives only • Offer management training and support systems to senior people only	<u>Systems</u> • Depend on many performance information system, providing data on customers especially • Distribute performance data widely • Offer management training and support systems to many people
<u>Culture</u> • Inwardly focused • Centralized • Slow to make decisions • Political • Risk averse	<u>Culture</u> • Externally oriented • Empowering • Quick to make decisions • Open and candid • More risk tolerant

1.9 Paradigm Shifts

From	To
Industrial Society	Information Society
Forced Technology	High Tech/High Touch
National Economy	World Economy
Short Term	Long Term
Centralization	Decentralization
Institutional Help	Self-Help
North	South
Either/OR	Multiple Option

References:

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† Timothy J. Kloppenborg, Vittal Anantatmula, Kathryn N. Wells, *“Contemporary Project Management”* 4th Edition, Cengage Learning.

